

Assignment 1 GenAI Analysis

For this assignment, the two generative AI programs used in this assignment were ChatGPT.com and Claude.ai. They were accessed through a Google query.

Identical prompts were given to both programs. The prompt read:

Write a C program that picks a secret number between 1 and 10. The user gets up to 3 tries to guess the number. Requirements: 1. Use a fixed secret number (for example, 7). 2. Use printf to prompt the user and scanf to read their guess. 3. Use a while or for loop to let the user guess up to 3 times. 4. Use if-else statements to check if the guess is too high, too low, or correct. 5. Stop the loop immediately if the user guesses the correct number. 6. Print a final win or lose message.

Example output: Guess a number between 1 and 10. Attempt 1/3. Enter your guess: 5 Too low! Try again. Attempt 2/3. Enter your guess: 8 Too high! Try again. Attempt 3/3. Enter your guess: 7 Correct! You win!

ChatGPT Generated Code:

```
#include <stdio.h>
```

```
int main() {
```

```
    int secretNumber = 7;
```

```
    int guess;
```

```
    int won = 0;
```

```
    printf("Guess a number between 1 and 10.\n");
```

```
    for (int attempt = 1; attempt <= 3; attempt++) {
```

```
        printf("Attempt %d/3.\n", attempt);
```

```
        printf("Enter your guess: ");
```

```
        scanf("%d", &guess);
```

```
if (guess == secretNumber) {

    printf("Correct! You win!\n");

    won = 1;

    break; // Stop immediately if correct

} else if (guess < secretNumber) {

    printf("Too low! Try again.\n");

} else {

    printf("Too high! Try again.\n");

}

}

if (!won) {

    printf("Sorry, you lose! The secret number was %d.\n", secretNumber);

}

return 0;

}
```

Claude.ai Generated Code:

```
#include <stdio.h>

int main() {

    int secretNumber = 7; // Fixed secret number

    int guess;

    int attempts = 3;

    int won = 0; // Flag to track if user guessed correctly

    printf("Guess a number between 1 and 10.\n");

    for (int i = 1; i <= attempts; i++) {

        printf("Attempt %d/%d. Enter your guess: ", i, attempts);

        scanf("%d", &guess);

        if (guess == secretNumber) {

            printf("Correct! You win!\n");
```

```
won = 1;

break; // Stop the loop immediately on correct guess

} else if (guess < secretNumber) {

    printf("Too low! Try again.\n");

} else {

    printf("Too high! Try again.\n");

}

}

if (!won) {

    printf("Sorry, you lose! The number was %d.\n", secretNumber);

}

return 0;

}
```

Correctness Analysis:

For ChatGPT the code does function and handles the sample input. The code cannot handle any edge cases such as strings or floats. The code crashes if a string or float is entered.

For Claude.ai the code does function and handle the sample input. The code cannot handle any edge cases such as strings or floats. The code crashes if a string or float is entered.

Time Complexity / Execution Time Analysis:

Comparing ChatGPT and Claude.ai, Claude.ai has the longer execution time due to its use of an extra variable called “attempts”. At the start of Claude’s code, it declares the attempts variable, which is used in its for loop. The for loop initializes a temporary variable i and checks if it is below or equal to the attempts variable value. This requires the program retrieve another stored value from the register for every iteration of the loop, increasing its execution time to be greater than ChatGPT’s, which simply checks if the temporary variable in the for loop is less than or equal to the constant 3.

Space Complexity Analysis:

Comparing ChatGPT and Claude.ai, Claude.ai has the greater space complexity. This is mostly due to its use of four stored variables, which intuitively uses up more memory and thereby more space. The program's use of the fourth variable, called "attempts", also increases space complexity by requiring the code to draw its value from the register for every iteration in the for loop.

Maintainability Analysis:

Neither ChatGPT nor Claude.ai could handle edge cases, such as bad inputs, with both program's codes crashing. Claude.ai produced a more maintainable code with its comments which, while very sparse, do outmatch ChatGPT's single comment on the break statement.

For the basis of this assignment ChatGPT will be chosen for rework. It was chosen for its superior execution time and space complexity. While it does have worse maintainability, this will be remedied in the improvement phase.

Improvement Phase:

ChatGPT's code was improved in terms of correctness and maintainability. It was determined that the execution time and space complexity could not be easily improved. For correctness, the code has been altered to handle edge-cases, such as unexpected inputs, with the addition of a new else segment in the if-else block that informs the user of a bad input. This bad input still counts as an attempt at the problem. This improvement to the code also assists its maintainability. In addition, thorough comments were added to the code for maintainability.